

Node Structure Visualization

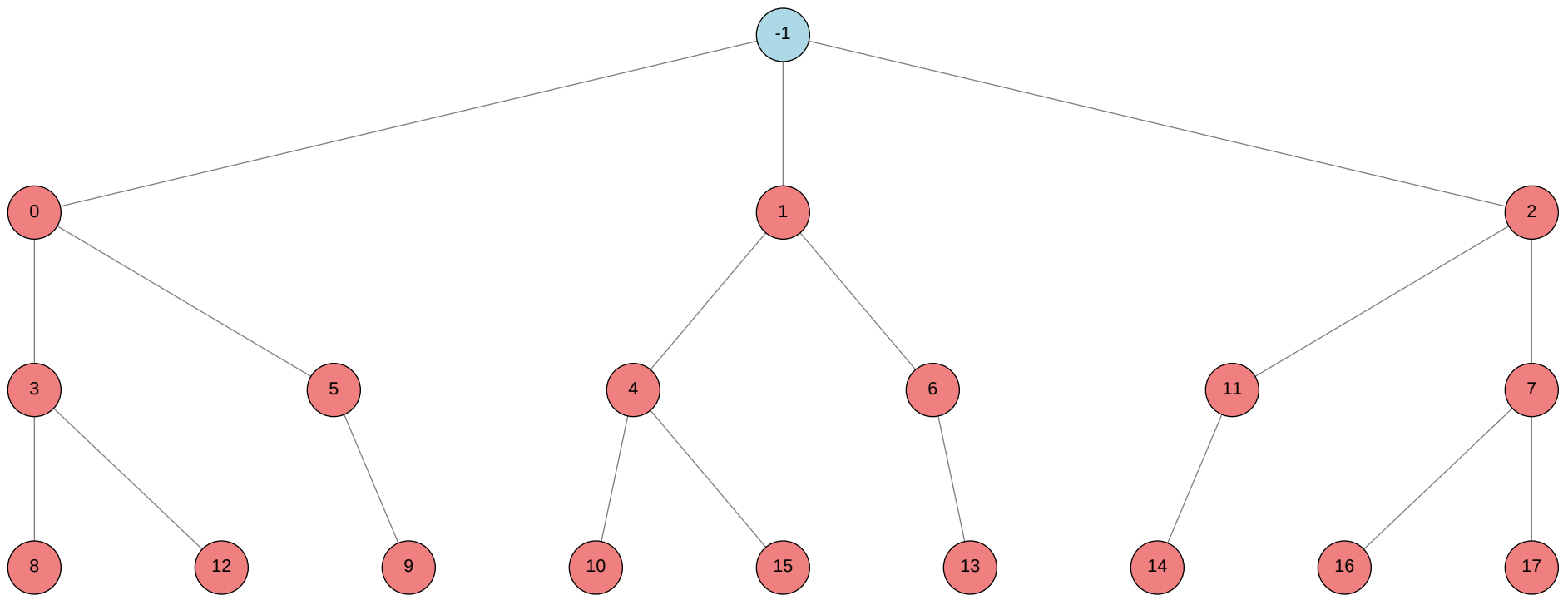
Click on any node in the tree to jump to its detailed information.

Total Nodes: 19 | No validation scores available

Node Tree:

Table of Contents:

Node ID	Stage	Status	Tool
>-1			
>0			autogluon.multimodal
>1			machine learning
>2			autogluon.tabular
>3			autogluon.multimodal
>4			machine learning
>5			autogluon.multimodal
>6			machine learning
>7			autogluon.tabular
>8			autogluon.multimodal
>9			autogluon.multimodal
>10			machine learning
>11			autogluon.tabular
>12			autogluon.multimodal
>13			machine learning
>14			autogluon.tabular
>15			machine learning
>16			autogluon.tabular
>17			autogluon.tabular



Status	Color
Success	Green
Failure	Red
Neutral	Blue

Node Details:

Node -1 (root)

Property	Value
uct_value	-0.4444
ctime	2026-05-21 16:14:31
debug_attempts	0
depth	0
execution_time	0.0
expected_child_count	0
failure_visits	18
id	-1
is_debug_successful	✗
is_leaf	✗
is_successful	✗
is_terminal	✓
num_children	3
prev_tutorial_prompt	None
stage	root
time_step	-1
tool_used	
tools_available	[]
unvalidated_visits	0
validated_reward	0.0
validated_visits	0
validation_score	None
visits	18
parent_id	None
child_ids	[0, 1, 2]

Node 0 (evolve)

Property	Value
uct_value	0.6481
ctime	2026-05-21 16:15:34
debug_attempts	2
depth	1
execution_time	0.0
expected_child_count	0
exploitation	-0.3
exploration	0.9968262051089475
failure_penalty	-0.3
failure_visits	6
id	0
is_debug_successful	✗
is_leaf	✗
is_successful	✗
is_terminal	✓
normalized_failure_visit	3
num_children	2
prev_tutorial_prompt	None
stage	evolve
time_step	0
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.0
validated_visits	0
validation_score	None

visits	6
parent_id	-1
child_ids	[3, 5]

Error Message:

stderr:

==> WARNING: A newer version of conda exists. <==

current version: 25.11.0

latest version: 26.5.0

Please update conda by running

\$ conda update -n base -c conda-forge conda

Using Python 3.11.15 environment at: 44-addition_2/node_0/conda_env

Resolved 235 packages in 1.54s

Uninstalled 1 package in 1.98s

Installed 233 packages in 1m 59s

+ absl-py==2.4.0

+ accelerate==1.13.0

+ adagio==0.2.6

+ aiohappyeyeballs==2.6.2

+ aiohttp==3.13.5

+ aiohttp-cors==0.8.1

+ aiosignal==1.4.0

+ alembic==1.18.4

+ annotated-doc==0.0.4

+ annotated-types==0.7.0

+ antlr4-python3-runtime==4.9.3

+ attrs==26.1.0

+ autogluon==1.5.1b20260521

+ autogluon-common==1.5.1b20260521

+ autogluon-core==1.5.1b20260521

+ autogluon-features==1.5.1b20260521

+ autogluon-multimodal==1.5.1b20260521

+ autogluon-tabular==1.5.1b20260521

+ autogluon-timeseries==1.5.1b20260521

+ beartype==0.22.9

+ beautifulsoup4==4.14.3

+ blis==1.3.3

+ boto3==1.43.12

+ botocore==1.43.12

+ catalogue==2.0.10

+ catboost==1.2.10

+ certifi==2026.5.20

+ cffi==2.0.0

- + chardet==7.4.3
- + charset-normalizer==3.4.7
- + chronos-forecasting==2.2.2
- + click==8.4.0
- + cloudpathlib==0.24.0
- + cloudpickle==3.1.2
- + colorama==0.4.6
- + colorful==0.6.0a1
- + colorlog==6.10.1
- + confection==1.3.3
- + contourpy==1.3.3
- + coreforecast==0.0.16
- + cryptography==48.0.0
- + cycler==0.12.1
- + cymem==2.0.13
- + datasets==4.0.0
- + defusedxml==0.7.1
- + dill==0.3.8
- + distlib==0.4.0
- + einops==0.8.2
- + einx==0.4.3
- + evaluate==0.4.6
- + fastai==2.8.5
- + fastcore==1.8.18
- + fastdownload==0.0.7
- + fastprogress==1.0.5
- + fasttransform==0.0.2
- + filelock==3.29.0
- + fonttools==4.63.0
- + frozendict==2.4.7
- + frozenlist==1.8.0
- + fsspec==2025.3.0
- + fugue==0.9.7
- + future==1.0.0
- + gdown==6.0.0
- + gluonts==0.16.2
- + google-api-core==2.30.3
- + google-auth==2.53.0
- + googleapis-common-protos==1.75.0
- + graphviz==0.21
- + grpcio==1.81.0rc1
- + hf-xet==1.5.0
- + httpx==1.0.dev3
- + huggingface-hub==0.36.2
- + hyperopt==0.2.7

+ idna==3.15
... [truncated, 6581 chars total]

Error Analysis:

ERROR_SUMMARY: The error occurs because the `fit()` method of AutoGluon's `MultiModalPredictor` does not accept an `eval_metric` keyword argument. Passing this unsupported argument results in a `TypeError`, preventing model training from starting.

SUGGESTED_FIX: Remove the `eval_metric="roc_auc"` argument from the `predictor.fit()` call. If you wish to specify a validation metric, use the `hyperparameters` dictionary (e.g., `"optimization.metric_name": "roc_auc"`) or rely on AutoGluon's default metric selection for binary classification. Check the [AutoGluon MultiModal documentation](<https://auto.gluon.ai/stable/tutorials/multimodal/index.html>) for the correct way to set evaluation metrics.

Node 1 (evolve)

Property	Value
uct_value	0.6481
ctime	2026-05-21 16:25:22
debug_attempts	2
depth	1
execution_time	0.0
expected_child_count	0
exploitation	-0.3
exploration	0.7804660679599289
failure_penalty	-0.3
failure_visits	6
id	1
is_debug_successful	✗
is_leaf	✗
is_successful	✗
is_terminal	✓
normalized_failure_visit	3
num_children	2
prev_tutorial_prompt	None
stage	evolve
time_step	1
tool_used	machine learning
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.0
validated_visits	0
validation_score	None

visits	6
parent_id	-1
child_ids	[4, 6]

Error Message:

stderr:

==> WARNING: A newer version of conda exists. <==

current version: 25.11.0

latest version: 26.5.0

Please update conda by running

\$ conda update -n base -c conda-forge conda

warning: Requirements file `~/home/anri21/.conda/envs/mlzero\_env/lib/python3.11/site-packages/autogluon/assistant/tools\_registry/machine learning/requirements.txt` does not contain any dependencies

Using Python 3.11.15 environment at: 44-addition_2/node_1/conda_env

Resolved 79 packages in 478ms

Uninstalled 1 package in 1.97s

Installed 78 packages in 44.80s

+ aiohappyeyeballs==2.6.2

+ aiohttp==3.13.5

+ aiosignal==1.4.0

+ annotated-doc==0.0.4

+ annotated-types==0.7.0

+ anyio==4.13.0

+ attrs==26.1.0

+ certifi==2026.5.20

+ chardet==7.4.3

+ charset-normalizer==3.4.7

+ click==8.4.0

+ cuda-bindings==13.2.0

+ cuda-pathfinder==1.5.4

+ cuda-toolkit==13.0.2

+ datasets==4.8.5

+ dill==0.4.1

+ filelock==3.29.0

+ frozenlist==1.8.0

+ fsspec==2026.2.0

+ h11==0.16.0

+ hf-xet==1.5.0

+ httpcore==1.0.9

+ httpx==0.28.1

+ huggingface-hub==1.16.0

+ idna==3.15

+ jinja2==3.1.6

- + markdown-it-py==4.2.0
- + markupsafe==3.0.3
- + mdurl==0.1.2
- + mpmath==1.3.0
- + multidict==6.7.1
- + multiprocessing==0.70.19
- + networkx==3.6.1
- + numpy==2.4.6
- + nvidia-cublas==13.1.1.3
- + nvidia-cuda-cupti==13.0.85
- + nvidia-cuda-nvrtc==13.0.88
- + nvidia-cuda-runtime==13.0.96
- + nvidia-cudnn-cu13==9.20.0.48
- + nvidia-cufft==12.0.0.61
- + nvidia-cufile==1.15.1.6
- + nvidia-curand==10.4.0.35
- + nvidia-cusolver==12.0.4.66
- + nvidia-cusparselt-cu13==0.8.1
- + nvidia-cusparse==12.6.3.3
- + nvidia-nccl-cu13==2.29.7
- + nvidia-nvjitlink==13.0.88
- + nvidia-nvshmem-cu13==3.4.5
- + nvidia-nvtx==13.0.85
- + opencv-python==4.13.0.92
- + pandas==3.0.3
- + pillow==12.2.0
- + propcache==0.5.2
- + pyarrow==24.0.0
- + pydantic==2.13.4
- + pydantic-core==2.46.4
- + pygments==2.20.0
- + python-calamine==0.6.2
- + python-dateutil==2.9.0.post0
- + pytz==2026.2
- + pyyaml==6.0.3
- + requests==2.34.2
- + rich==15.0.0
- setuptools==82.0.1
- + set... [truncated, 7648 chars total]

Error Analysis:

ERROR_SUMMARY: The error "DNN library initialization failed" (CUDNN_STATUS_NOT_INITIALIZED) occurs during image feature extraction with TensorFlow/Keras EfficientNetB0. This is caused by a mismatch or incompatibility between the installed NVIDIA driver, CUDA toolkit, and cuDNN library versions, or by improper GPU environment setup (e.g., insufficient driver version, missing libraries, or lack of permissions).

SUGGESTED_FIX:

Check that the NVIDIA driver version is compatible with the installed CUDA and cuDNN versions (driver 565.57.1 may be too old or mismatched for CUDA 12.5/cuDNN 9.20). Ensure that the correct CUDA and cuDNN libraries are installed and visible to TensorFlow, and that the GPU is not in use by other processes or out of memory. If running in a managed environment, consult system documentation or administrators to update drivers/libraries or try running with CPU-only TensorFlow as a temporary workaround.

Node 2 (evolve)

Property	Value
uct_value	0.6481
ctime	2026-05-21 16:36:04
debug_attempts	2
depth	1
execution_time	0.0
expected_child_count	0
exploitation	-0.3
exploration	0.53219899601806
failure_penalty	-0.3
failure_visits	6
id	2
is_debug_successful	✗
is_leaf	✗
is_successful	✗
is_terminal	✓
normalized_failure_visit	3
num_children	2
prev_tutorial_prompt	None
stage	evolve
time_step	2
tool_used	autogluon.tabular
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.0
validated_visits	0
validation_score	None

visits	6
parent_id	-1
child_ids	[11, 7]

Error Message:

stderr:

==> WARNING: A newer version of conda exists. <==

current version: 25.11.0

latest version: 26.5.0

Please update conda by running

\$ conda update -n base -c conda-forge conda

Using Python 3.11.15 environment at: 44-addition_2/node_2/conda_env

Resolved 239 packages in 1.43s

Uninstalled 1 package in 2.02s

Installed 237 packages in 1m 50s

+ absl-py==2.4.0

+ accelerate==1.13.0

+ adagio==0.2.6

+ aiohappyeyeballs==2.6.2

+ aiohttp==3.13.5

+ aiohttp-cors==0.8.1

+ aiosignal==1.4.0

+ alembic==1.18.4

+ annotated-doc==0.0.4

+ annotated-types==0.7.0

+ antlr4-python3-runtime==4.9.3

+ attrs==26.1.0

+ autogluon==1.5.1b20260521

+ autogluon-common==1.5.1b20260521

+ autogluon-core==1.5.1b20260521

+ autogluon-features==1.5.1b20260521

+ autogluon-multimodal==1.5.1b20260521

+ autogluon-tabular==1.5.1b20260521

+ autogluon-timeseries==1.5.1b20260521

+ beartype==0.22.9

+ beautifulsoup4==4.14.3

+ blis==1.3.3

+ boto3==1.43.12

+ botocore==1.43.12

+ catalogue==2.0.10

+ catboost==1.2.10

+ certifi==2026.5.20

+ cffi==2.0.0

- + chardet==7.4.3
- + charset-normalizer==3.4.7
- + chronos-forecasting==2.2.2
- + click==8.4.0
- + cloudpathlib==0.24.0
- + cloudpickle==3.1.2
- + colorama==0.4.6
- + colorful==0.6.0a1
- + colorlog==6.10.1
- + confection==1.3.3
- + contourpy==1.3.3
- + coreforecast==0.0.16
- + cryptography==48.0.0
- + cycler==0.12.1
- + cymem==2.0.13
- + daal==2025.9.0
- + datasets==4.0.0
- + defusedxml==0.7.1
- + dill==0.3.8
- + distlib==0.4.0
- + einops==0.8.2
- + einx==0.4.3
- + evaluate==0.4.6
- + fastai==2.8.5
- + fastcore==1.8.18
- + fastdownload==0.0.7
- + fastprogress==1.0.5
- + fasttransform==0.0.2
- + filelock==3.29.0
- + fonttools==4.63.0
- + frozendict==2.4.7
- + frozenlist==1.8.0
- + fsspec==2025.3.0
- + fugue==0.9.7
- + future==1.0.0
- + gdown==6.0.0
- + gluonts==0.16.2
- + google-api-core==2.30.3
- + google-auth==2.53.0
- + googleapis-common-protos==1.75.0
- + graphviz==0.21
- + grpcio==1.81.0rc1
- + hf-xet==1.5.0
- + httpx==1.0.dev3
- + huggingface-hub==0.36.2

+ hyperopt==0.... [truncated, 174471 chars total]

Error Analysis:

ERROR_SUMMARY: The error occurs because the trained AutoGluon Tabular model expects the columns 'skin_tone', 'alternative_skin_tone', and 'expert_opinion' in the test data, but the provided test set only contains 'image_name'. As a result, a KeyError is raised during prediction since the required features are missing from the test input.

SUGGESTED_FIX:

Switch to a multimodal approach using AutoGluon MultimodalPredictor or TabularPredictor with multimodal hyperparameters, so that the model can learn directly from image files (using the 'image_name' column as an image path) rather than relying on tabular metadata that is unavailable at test time. Update your training and prediction pipeline to treat 'image_name' as an image feature, configure feature metadata accordingly, and ensure that image file paths are correctly resolved for both training and test sets.

Node 3 (debug)

Property	Value
uct_value	0.9261
ctime	2026-05-21 16:46:29
debug_attempts	3
depth	2
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	1.2684446626418298
failure_penalty	-0.0
failure_visits	3
id	3
is_debug_successful	✗
is_leaf	✗
is_successful	✗
is_terminal	✓
normalized_failure_visit	0
num_children	2
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	debug
time_step	3
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.0

validated_visits	0
validation_score	None
visits	3
parent_id	0
child_ids	[8, 12]

Error Message:

stderr:

==> WARNING: A newer version of conda exists. <==

current version: 25.11.0

latest version: 26.5.0

Please update conda by running

\$ conda update -n base -c conda-forge conda

Using Python 3.11.15 environment at: 44-addition_2/node_3/conda_env

Resolved 235 packages in 1.44s

Uninstalled 1 package in 1.89s

Installed 233 packages in 1m 41s

+ absl-py==2.4.0

+ accelerate==1.13.0

+ adagio==0.2.6

+ aiohappyeyeballs==2.6.2

+ aiohttp==3.13.5

+ aiohttp-cors==0.8.1

+ aiosignal==1.4.0

+ alembic==1.18.4

+ annotated-doc==0.0.4

+ annotated-types==0.7.0

+ antlr4-python3-runtime==4.9.3

+ attrs==26.1.0

+ autogluon==1.5.1b20260521

+ autogluon-common==1.5.1b20260521

+ autogluon-core==1.5.1b20260521

+ autogluon-features==1.5.1b20260521

+ autogluon-multimodal==1.5.1b20260521

+ autogluon-tabular==1.5.1b20260521

+ autogluon-timeseries==1.5.1b20260521

+ beartype==0.22.9

+ beautifulsoup4==4.14.3

+ blis==1.3.3

+ boto3==1.43.12

+ botocore==1.43.12

+ catalogue==2.0.10

- + catboost==1.2.10
- + certifi==2026.5.20
- + cffi==2.0.0
- + chardet==7.4.3
- + charset-normalizer==3.4.7
- + chronos-forecasting==2.2.2
- + click==8.4.0
- + cloudpathlib==0.24.0
- + cloudpickle==3.1.2
- + colorama==0.4.6
- + colorful==0.6.0a1
- + colorlog==6.10.1
- + confection==1.3.3
- + contourpy==1.3.3
- + coreforecast==0.0.16
- + cryptography==48.0.0
- + cycler==0.12.1
- + cymem==2.0.13
- + datasets==4.0.0
- + defusedxml==0.7.1
- + dill==0.3.8
- + distlib==0.4.0
- + einops==0.8.2
- + einx==0.4.3
- + evaluate==0.4.6
- + fastai==2.8.5
- + fastcore==1.8.18
- + fastdownload==0.0.7
- + fastprogress==1.0.5
- + fasttransform==0.0.2
- + filelock==3.29.0
- + fonttools==4.63.0
- + frozendict==2.4.7
- + frozenlist==1.8.0
- + fsspec==2025.3.0
- + fugue==0.9.7
- + future==1.0.0
- + gdown==6.0.0
- + gluonts==0.16.2
- + google-api-core==2.30.3
- + google-auth==2.53.0
- + googleapis-common-protos==1.75.0
- + graphviz==0.21
- + grpcio==1.81.0rc1
- + hf-xet==1.5.0

+ httpx==1.0.dev3
+ huggingface-hub==0.36.2
+ hyperopt==0.2.7
+ idna==3.15
... [truncated, 9959 chars total]

Error Analysis:

ERROR_SUMMARY: The script failed during model training because the hyperparameter dictionary passed to AutoGluon's MultiModalPredictor.fit() included the key 'optimization.metric_name', which is not recognized. The correct key is 'optim.metric_name', and using the deprecated or incorrect prefix caused a KeyError, halting execution before any predictions or validation could be produced.

SUGGESTED_FIX: Update the hyperparameters dictionary in your predictor.fit() call by replacing 'optimization.metric_name' with 'optim.metric_name'. Double-check all hyperparameter keys for correct and current naming conventions as per the latest AutoGluon documentation, then rerun the script.

Node 4 (debug)

Property	Value
uct_value	0.9261
ctime	2026-05-21 16:55:07
debug_attempts	3
depth	2
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	1.2684446626418298
failure_penalty	-0.0
failure_visits	3
id	4
is_debug_successful	✗
is_leaf	✗
is_successful	✗
is_terminal	✓
normalized_failure_visit	0
num_children	2
prev_tutorial_prompt	### Condensed: Intro to Use Machine Learning or Deep Learning algorithms # Condensed: Intro to Use Machine Learning or Deep Learning algorithms Summary: You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you # Intro to Use Machine Learning or Deep Learning algorithms You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you have then
stage	debug
time_step	4
tool_used	machine learning

tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.0
validated_visits	0
validation_score	None
visits	3
parent_id	1
child_ids	[10, 15]

Error Message:

```

stderr:
==> WARNING: A newer version of conda exists. <==
current version: 25.11.0
latest version: 26.5.0
Please update conda by running
$ conda update -n base -c conda-forge conda
warning: Requirements file `~/home/anri21/.conda/envs/mlzero_env/lib/python3.11/site-packages/autogluon/assistant/tools_registry/machine learning/requirements.txt`
does not contain any dependencies
Using Python 3.11.15 environment at: 44-addition_2/node_4/conda_env
Resolved 79 packages in 366ms
Uninstalled 1 package in 1.88s
Installed 78 packages in 39.19s
+ aiohappyeyeballs==2.6.2
+ aiohttp==3.13.5
+ aiosignal==1.4.0
+ annotated-doc==0.0.4
+ annotated-types==0.7.0
+ anyio==4.13.0
+ attrs==26.1.0
+ certifi==2026.5.20
+ chardet==7.4.3
+ charset-normalizer==3.4.7
+ click==8.4.0
+ cuda-bindings==13.2.0
+ cuda-pathfinder==1.5.4
+ cuda-toolkit==13.0.2
+ datasets==4.8.5
+ dill==0.4.1
+ filelock==3.29.0

```

- + frozenlist==1.8.0
- + fsspec==2026.2.0
- + h11==0.16.0
- + hf-xet==1.5.0
- + httpcore==1.0.9
- + httpx==0.28.1
- + huggingface-hub==1.16.0
- + idna==3.15
- + jinja2==3.1.6
- + markdown-it-py==4.2.0
- + markupsafe==3.0.3
- + mdurl==0.1.2
- + mpmath==1.3.0
- + multidict==6.7.1
- + multiprocessing==0.70.19
- + networkx==3.6.1
- + numpy==2.4.6
- + nvidia-cublas==13.1.1.3
- + nvidia-cuda-cupti==13.0.85
- + nvidia-cuda-nvrtc==13.0.88
- + nvidia-cuda-runtime==13.0.96
- + nvidia-cudnn-cu13==9.20.0.48
- + nvidia-cufft==12.0.0.61
- + nvidia-cufile==1.15.1.6
- + nvidia-curand==10.4.0.35
- + nvidia-cusolver==12.0.4.66
- + nvidia-cusparselt-cu13==0.8.1
- + nvidia-cusparse==12.6.3.3
- + nvidia-nccl-cu13==2.29.7
- + nvidia-nvjitlink==13.0.88
- + nvidia-nvshmem-cu13==3.4.5
- + nvidia-nvtx==13.0.85
- + opencv-python==4.13.0.92
- + pandas==3.0.3
- + pillow==12.2.0
- + propcache==0.5.2
- + pyarrow==24.0.0
- + pydantic==2.13.4
- + pydantic-core==2.46.4
- + pygments==2.20.0
- + python-calamine==0.6.2
- + python-dateutil==2.9.0.post0
- + pytz==2026.2
- + pyyaml==6.0.3
- + requests==2.34.2

```
+ rich==15.0.0
- setuptools==82.0.1
+ set... [truncated, 7626 chars total]
```

Error Analysis:

ERROR_SUMMARY: TensorFlow failed to initialize the cuDNN DNN library on the GPU, resulting in a runtime error during image feature extraction with EfficientNetB0. The error log indicates "DNN library initialization failed" and "Possibly insufficient driver version: 565.57.1", suggesting a mismatch or incompatibility between the installed NVIDIA driver, CUDA toolkit, and cuDNN versions required by TensorFlow.

SUGGESTED_FIX:

Check that your NVIDIA driver, CUDA toolkit, and cuDNN versions are all compatible with the version of TensorFlow you are using (see TensorFlow's official GPU support matrix). If necessary, update the NVIDIA driver to a version recommended for your CUDA/cuDNN stack, and ensure that the correct CUDA and cuDNN libraries are visible in your environment (e.g., via LD_LIBRARY_PATH). After making these changes, restart your environment and rerun the script to verify that TensorFlow can successfully initialize the GPU and cuDNN.

Node 5 (debug)

Property	Value
uct_value	1.3384
ctime	2026-05-21 17:04:56
debug_attempts	3
depth	2
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	1.664857771836881
failure_penalty	-0.0
failure_visits	2
id	5
is_debug_successful	✗
is_leaf	✗
is_successful	✗
is_terminal	✓
normalized_failure_visit	0
num_children	1
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	debug
time_step	5
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.0

validated_visits	0
validation_score	None
visits	2
parent_id	0
child_ids	[9]

Error Message:

stderr:

==> WARNING: A newer version of conda exists. <==

current version: 25.11.0

latest version: 26.5.0

Please update conda by running

\$ conda update -n base -c conda-forge conda

Using Python 3.11.15 environment at: 44-addition_2/node_5/conda_env

Resolved 235 packages in 3.66s

Uninstalled 1 package in 2.38s

Installed 233 packages in 2m 03s

+ absl-py==2.4.0

+ accelerate==1.13.0

+ adagio==0.2.6

+ aiohappyeyeballs==2.6.2

+ aiohttp==3.13.5

+ aiohttp-cors==0.8.1

+ aiosignal==1.4.0

+ alembic==1.18.4

+ annotated-doc==0.0.4

+ annotated-types==0.7.0

+ antlr4-python3-runtime==4.9.3

+ attrs==26.1.0

+ autogluon==1.5.1b20260521

+ autogluon-common==1.5.1b20260521

+ autogluon-core==1.5.1b20260521

+ autogluon-features==1.5.1b20260521

+ autogluon-multimodal==1.5.1b20260521

+ autogluon-tabular==1.5.1b20260521

+ autogluon-timeseries==1.5.1b20260521

+ beartype==0.22.9

+ beautifulsoup4==4.14.3

+ blis==1.3.3

+ boto3==1.43.12

+ botocore==1.43.12

+ catalogue==2.0.10

- + catboost==1.2.10
- + certifi==2026.5.20
- + cffi==2.0.0
- + chardet==7.4.3
- + charset-normalizer==3.4.7
- + chronos-forecasting==2.2.2
- + click==8.4.0
- + cloudpathlib==0.24.0
- + cloudpickle==3.1.2
- + colorama==0.4.6
- + colorful==0.6.0a1
- + colorlog==6.10.1
- + confection==1.3.3
- + contourpy==1.3.3
- + coreforecast==0.0.16
- + cryptography==48.0.0
- + cycler==0.12.1
- + cymem==2.0.13
- + datasets==4.0.0
- + defusedxml==0.7.1
- + dill==0.3.8
- + distlib==0.4.0
- + einops==0.8.2
- + einx==0.4.3
- + evaluate==0.4.6
- + fastai==2.8.5
- + fastcore==1.8.18
- + fastdownload==0.0.7
- + fastprogress==1.0.5
- + fasttransform==0.0.2
- + filelock==3.29.0
- + fonttools==4.63.0
- + frozendict==2.4.7
- + frozenlist==1.8.0
- + fsspec==2025.3.0
- + fugue==0.9.7
- + future==1.0.0
- + gdown==6.0.0
- + gluonts==0.16.2
- + google-api-core==2.30.3
- + google-auth==2.53.0
- + googleapis-common-protos==1.75.0
- + graphviz==0.21
- + grpcio==1.81.0rc1
- + hf-xet==1.5.0

```
+ httpx==1.0.dev3
+ huggingface-hub==0.36.2
+ hyperopt==0.2.7
+ idna==3.15
... [truncated, 9718 chars total]
```

Error Analysis:

ERROR_SUMMARY: The error occurs because the hyperparameter `optim.metric\_name` is not recognized in the AutoGluon MultiModalPredictor configuration. This parameter does not exist in the expected config schema, causing a KeyError during model training when overrides are applied.

SUGGESTED_FIX:

Remove the `"optim.metric\_name": "roc\_auc"` entry from the `hyperparameters` dictionary in your `predictor.fit()` call. AutoGluon MultiModalPredictor does not support this parameter; instead, rely on its default metric selection for binary classification or consult the documentation for supported metric configuration options.

Node 6 (debug)

Property	Value
uct_value	1.3384
ctime	2026-05-21 17:14:25
debug_attempts	3
depth	2
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	1.664857771836881
failure_penalty	-0.0
failure_visits	2
id	6
is_debug_successful	✗
is_leaf	✗
is_successful	✗
is_terminal	✓
normalized_failure_visit	0
num_children	1
prev_tutorial_prompt	### Condensed: Intro to Use Machine Learning or Deep Learning algorithms # Condensed: Intro to Use Machine Learning or Deep Learning algorithms Summary: You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you # Intro to Use Machine Learning or Deep Learning algorithms You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you have then
stage	debug
time_step	6
tool_used	machine learning

tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.0
validated_visits	0
validation_score	None
visits	2
parent_id	1
child_ids	[13]

Error Message:

```

stderr:
==> WARNING: A newer version of conda exists. <==
current version: 25.11.0
latest version: 26.5.0
Please update conda by running
$ conda update -n base -c conda-forge conda
warning: Requirements file `~/home/anri21/.conda/envs/mlzero_env/lib/python3.11/site-packages/autogluon/assistant/tools_registry/machine learning/requirements.txt`
does not contain any dependencies
Using Python 3.11.15 environment at: 44-addition_2/node_6/conda_env
Resolved 79 packages in 643ms
Uninstalled 1 package in 1.89s
Installed 78 packages in 38.48s
+ aiohappyeyeballs==2.6.2
+ aiohttp==3.13.5
+ aiosignal==1.4.0
+ annotated-doc==0.0.4
+ annotated-types==0.7.0
+ anyio==4.13.0
+ attrs==26.1.0
+ certifi==2026.5.20
+ chardet==7.4.3
+ charset-normalizer==3.4.7
+ click==8.4.0
+ cuda-bindings==13.2.0
+ cuda-pathfinder==1.5.4
+ cuda-toolkit==13.0.2
+ datasets==4.8.5
+ dill==0.4.1
+ filelock==3.29.0

```

- + frozenlist==1.8.0
- + fsspec==2026.2.0
- + h11==0.16.0
- + hf-xet==1.5.0
- + httpcore==1.0.9
- + httpx==0.28.1
- + huggingface-hub==1.16.0
- + idna==3.15
- + jinja2==3.1.6
- + markdown-it-py==4.2.0
- + markupsafe==3.0.3
- + mdurl==0.1.2
- + mpmath==1.3.0
- + multidict==6.7.1
- + multiprocessing==0.70.19
- + networkx==3.6.1
- + numpy==2.4.6
- + nvidia-cublas==13.1.1.3
- + nvidia-cuda-cupti==13.0.85
- + nvidia-cuda-nvrtc==13.0.88
- + nvidia-cuda-runtime==13.0.96
- + nvidia-cudnn-cu13==9.20.0.48
- + nvidia-cufft==12.0.0.61
- + nvidia-cufile==1.15.1.6
- + nvidia-curand==10.4.0.35
- + nvidia-cusolver==12.0.4.66
- + nvidia-cusparselt-cu13==0.8.1
- + nvidia-cusparse==12.6.3.3
- + nvidia-nccl-cu13==2.29.7
- + nvidia-nvjitlink==13.0.88
- + nvidia-nvshmem-cu13==3.4.5
- + nvidia-nvtx==13.0.85
- + opencv-python==4.13.0.92
- + pandas==3.0.3
- + pillow==12.2.0
- + propcache==0.5.2
- + pyarrow==24.0.0
- + pydantic==2.13.4
- + pydantic-core==2.46.4
- + pygments==2.20.0
- + python-calamine==0.6.2
- + python-dateutil==2.9.0.post0
- + pytz==2026.2
- + pyyaml==6.0.3
- + requests==2.34.2

+ rich==15.0.0
- setuptools==82.0.1
+ set... [truncated, 7712 chars total]

Error Analysis:

ERROR_SUMMARY: TensorFlow failed to initialize the GPU DNN library (cuDNN), resulting in a runtime error during image feature extraction with EfficientNetB0. The error log shows "Could not create cudnn handle: CUDNN_STATUS_NOT_INITIALIZED" and "DNN library initialization failed," which typically indicates a mismatch between the installed CUDA/cuDNN versions and the NVIDIA driver, or insufficient available GPU memory/resources.

SUGGESTED_FIX:

Check that your NVIDIA driver version is fully compatible with the installed CUDA and cuDNN versions (driver 565.57.1 is reported, but TensorFlow and cuDNN may expect a different version). Ensure that the CUDA toolkit, cuDNN, and TensorFlow are all built for the same CUDA version, and that there is sufficient free GPU memory before running the script. If running in a shared environment, try restarting the runtime or node to clear GPU memory, and consider downgrading or upgrading TensorFlow/CUDA/cuDNN to a set of versions known to work together with your driver and hardware.

Node 7 (debug)

Property	Value
uct_value	0.9261
ctime	2026-05-21 17:24:05
debug_attempts	3
depth	2
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	1.2684446626418298
failure_penalty	-0.0
failure_visits	3
id	7
is_debug_successful	✗
is_leaf	✗
is_successful	✗
is_terminal	✓
normalized_failure_visit	0
num_children	2
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates using AutoGluon Multimodal for multimodal classification on the PetFinder dataset.
stage	debug
time_step	7
tool_used	autogluon.tabular
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.0

validated_visits	0
validation_score	None
visits	3
parent_id	2
child_ids	[16, 17]

Error Message:

stderr:

==> WARNING: A newer version of conda exists. <==

current version: 25.11.0

latest version: 26.5.0

Please update conda by running

\$ conda update -n base -c conda-forge conda

Using Python 3.11.15 environment at: 44-addition_2/node_7/conda_env

Resolved 239 packages in 1.51s

Uninstalled 1 package in 1.83s

Installed 237 packages in 1m 36s

+ absl-py==2.4.0

+ accelerate==1.13.0

+ adagio==0.2.6

+ aiohappyeyeballs==2.6.2

+ aiohttp==3.13.5

+ aiohttp-cors==0.8.1

+ aiosignal==1.4.0

+ alembic==1.18.4

+ annotated-doc==0.0.4

+ annotated-types==0.7.0

+ antlr4-python3-runtime==4.9.3

+ attrs==26.1.0

+ autogluon==1.5.1b20260521

+ autogluon-common==1.5.1b20260521

+ autogluon-core==1.5.1b20260521

+ autogluon-features==1.5.1b20260521

+ autogluon-multimodal==1.5.1b20260521

+ autogluon-tabular==1.5.1b20260521

+ autogluon-timeseries==1.5.1b20260521

+ beartype==0.22.9

+ beautifulsoup4==4.14.3

+ blis==1.3.3

+ boto3==1.43.12

+ botocore==1.43.12

+ catalogue==2.0.10

- + catboost==1.2.10
- + certifi==2026.5.20
- + cffi==2.0.0
- + chardet==7.4.3
- + charset-normalizer==3.4.7
- + chronos-forecasting==2.2.2
- + click==8.4.0
- + cloudpathlib==0.24.0
- + cloudpickle==3.1.2
- + colorama==0.4.6
- + colorful==0.6.0a1
- + colorlog==6.10.1
- + confection==1.3.3
- + contourpy==1.3.3
- + coreforecast==0.0.16
- + cryptography==48.0.0
- + cycler==0.12.1
- + cymem==2.0.13
- + daal==2025.9.0
- + datasets==4.0.0
- + defusedxml==0.7.1
- + dill==0.3.8
- + distlib==0.4.0
- + einops==0.8.2
- + einx==0.4.3
- + evaluate==0.4.6
- + fastai==2.8.5
- + fastcore==1.8.18
- + fastdownload==0.0.7
- + fastprogress==1.0.5
- + fasttransform==0.0.2
- + filelock==3.29.0
- + fonttools==4.63.0
- + frozendict==2.4.7
- + frozenlist==1.8.0
- + fsspec==2025.3.0
- + fugue==0.9.7
- + future==1.0.0
- + gdown==6.0.0
- + gluonts==0.16.2
- + google-api-core==2.30.3
- + google-auth==2.53.0
- + googleapis-common-protos==1.75.0
- + graphviz==0.21
- + grpcio==1.81.0rc1

```
+ hf-xet==1.5.0
+ httpx==1.0.dev3
+ huggingface-hub==0.36.2
+ hyperopt==0.... [truncated, 173922 chars total]
```

Error Analysis:

ERROR_SUMMARY: The error occurs because the test set provided to the trained AutoGluon TabularPredictor contains only the 'image_name' column, while the model expects the same feature columns as used during training: 'skin_tone', 'alternative_skin_tone', and 'expert_opinion'. This column mismatch leads to a KeyError when the predictor attempts to transform the test data, as the required columns are missing.

SUGGESTED_FIX:

Modify your inference pipeline to ensure that the test set passed to ``predictor.predict_proba()`` contains all the feature columns used during training ('skin_tone', 'alternative_skin_tone', 'expert_opinion'), not just 'image_name'. If these features are not available for the test set, you must either (a) generate or impute reasonable values for them, or (b) retrain your model using only features that are present in both train and test (e.g., using only image data via a multimodal approach).

Node 8 (debug)

Property	Value
uct_value	1.4821
ctime	2026-05-21 17:33:39
debug_attempts	2
depth	3
execution_time	0.0
expected_child_count	0
failure_visits	1
id	8
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✓
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	debug
time_step	8
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.0
validated_visits	0
validation_score	None
visits	1
parent_id	3
child_ids	[]

Error Message:

stderr:

==> WARNING: A newer version of conda exists. <==

current version: 25.11.0

latest version: 26.5.0

Please update conda by running

\$ conda update -n base -c conda-forge conda

Using Python 3.11.15 environment at: 44-addition_2/node_8/conda_env

Resolved 235 packages in 1.76s

Uninstalled 1 package in 1.92s

Installed 233 packages in 1m 24s

+ absl-py==2.4.0

+ accelerate==1.13.0

+ adagio==0.2.6

+ aiohappyeyeballs==2.6.2

+ aiohttp==3.13.5

+ aiohttp-cors==0.8.1

+ aiosignal==1.4.0

+ alembic==1.18.4

+ annotated-doc==0.0.4

+ annotated-types==0.7.0

+ antlr4-python3-runtime==4.9.3

+ attrs==26.1.0

+ autogluon==1.5.1b20260521

+ autogluon-common==1.5.1b20260521

+ autogluon-core==1.5.1b20260521

+ autogluon-features==1.5.1b20260521

+ autogluon-multimodal==1.5.1b20260521

+ autogluon-tabular==1.5.1b20260521

+ autogluon-timeseries==1.5.1b20260521

+ beartype==0.22.9

+ beautifulsoup4==4.14.3

+ blis==1.3.3

+ boto3==1.43.12

+ botocore==1.43.12

+ catalogue==2.0.10

+ catboost==1.2.10

+ certifi==2026.5.20

+ cffi==2.0.0

+ chardet==7.4.3

+ charset-normalizer==3.4.7

+ chronos-forecasting==2.2.2

+ click==8.4.0

+ cloudpathlib==0.24.0

+ cloudpickle==3.1.2
+ colorama==0.4.6
+ colorful==0.6.0a1
+ colorlog==6.10.1
+ confection==1.3.3
+ contourpy==1.3.3
+ coreforecast==0.0.16
+ cryptography==48.0.0
+ cycler==0.12.1
+ cymem==2.0.13
+ datasets==4.0.0
+ defusedxml==0.7.1
+ dill==0.3.8
+ distlib==0.4.0
+ einops==0.8.2
+ einx==0.4.3
+ evaluate==0.4.6
+ fastai==2.8.5
+ fastcore==1.8.18
+ fastdownload==0.0.7
+ fastprogress==1.0.5
+ fasttransform==0.0.2
+ filelock==3.29.0
+ fonttools==4.63.0
+ frozendict==2.4.7
+ frozenlist==1.8.0
+ fsspec==2025.3.0
+ fugue==0.9.7
+ future==1.0.0
+ gdown==6.0.0
+ gluonts==0.16.2
+ google-api-core==2.30.3
+ google-auth==2.53.0
+ googleapis-common-protos==1.75.0
+ graphviz==0.21
+ grpcio==1.81.0rc1
+ hf-xet==1.5.0
+ httpx==1.0.dev3
+ huggingface-hub==0.36.2
+ hyperopt==0.2.7
+ idna==3.15
... [truncated, 14773 chars total]

Error Analysis:

ERROR_SUMMARY: The root cause of the error is a distributed training initialization failure in PyTorch Lightning, specifically a `torch.distributed.DistNetworkError` indicating that the server socket could not listen on the required port (20620) because the address was already in use (`EADDRINUSE`). This prevented the distributed data parallel (DDP) strategy from starting, causing the entire model training process to fail.

SUGGESTED_FIX:

Check if another process is already using port 20620 on the machine and terminate it, or configure your training script to use a different, free port for distributed training by setting the `MASTER_PORT` environment variable before running the script. Alternatively, if distributed training is not required (e.g., you only have one GPU), set the AutoGluon hyperparameter `"env.strategy": "auto"` or `"env.num_gpus": 1` to avoid DDP and use single-GPU training, which will bypass the port conflict.

Node 9 (debug)

Property	Value
uct_value	1.1772
ctime	2026-05-21 17:43:09
debug_attempts	3
depth	3
execution_time	0.0
expected_child_count	0
failure_visits	1
id	9
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✓
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	debug
time_step	9
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.0
validated_visits	0
validation_score	None
visits	1
parent_id	5
child_ids	[]

Error Message:

stderr:

==> WARNING: A newer version of conda exists. <==

current version: 25.11.0

latest version: 26.5.0

Please update conda by running

\$ conda update -n base -c conda-forge conda

Using Python 3.11.15 environment at: 44-addition_2/node_9/conda_env

Resolved 235 packages in 1.20s

Uninstalled 1 package in 1.74s

Installed 233 packages in 1m 34s

+ absl-py==2.4.0

+ accelerate==1.13.0

+ adagio==0.2.6

+ aiohappyeyeballs==2.6.2

+ aiohttp==3.13.5

+ aiohttp-cors==0.8.1

+ aiosignal==1.4.0

+ alembic==1.18.4

+ annotated-doc==0.0.4

+ annotated-types==0.7.0

+ antlr4-python3-runtime==4.9.3

+ attrs==26.1.0

+ autogluon==1.5.1b20260521

+ autogluon-common==1.5.1b20260521

+ autogluon-core==1.5.1b20260521

+ autogluon-features==1.5.1b20260521

+ autogluon-multimodal==1.5.1b20260521

+ autogluon-tabular==1.5.1b20260521

+ autogluon-timeseries==1.5.1b20260521

+ beartype==0.22.9

+ beautifulsoup4==4.14.3

+ blis==1.3.3

+ boto3==1.43.12

+ botocore==1.43.12

+ catalogue==2.0.10

+ catboost==1.2.10

+ certifi==2026.5.20

+ cffi==2.0.0

+ chardet==7.4.3

+ charset-normalizer==3.4.7

+ chronos-forecasting==2.2.2

+ click==8.4.0

+ cloudpathlib==0.24.0

+ cloudpickle==3.1.2
+ colorama==0.4.6
+ colorful==0.6.0a1
+ colorlog==6.10.1
+ confection==1.3.3
+ contourpy==1.3.3
+ coreforecast==0.0.16
+ cryptography==48.0.0
+ cycler==0.12.1
+ cymem==2.0.13
+ datasets==4.0.0
+ defusedxml==0.7.1
+ dill==0.3.8
+ distlib==0.4.0
+ einops==0.8.2
+ einx==0.4.3
+ evaluate==0.4.6
+ fastai==2.8.5
+ fastcore==1.8.18
+ fastdownload==0.0.7
+ fastprogress==1.0.5
+ fasttransform==0.0.2
+ filelock==3.29.0
+ fonttools==4.63.0
+ frozendict==2.4.7
+ frozenlist==1.8.0
+ fsspec==2025.3.0
+ fugue==0.9.7
+ future==1.0.0
+ gdown==6.0.0
+ gluonts==0.16.2
+ google-api-core==2.30.3
+ google-auth==2.53.0
+ googleapis-common-protos==1.75.0
+ graphviz==0.21
+ grpcio==1.81.0rc1
+ hf-xet==1.5.0
+ httpx==1.0.dev3
+ huggingface-hub==0.36.2
+ hyperopt==0.2.7
+ idna==3.15
... [truncated, 14859 chars total]

Error Analysis:

ERROR_SUMMARY: The error occurs during distributed training initialization with PyTorch Lightning/DDP: `torch.distributed.DistNetworkError` ("address already in use" on port 20620). This means the process tried to start a distributed backend (for multi-GPU training) but the required network port was already occupied, likely by another process or a previous run that did not clean up properly.

SUGGESTED_FIX:

Set the `"env.num_gpus"` hyperparameter to `1` in your `predictor.fit()` call to disable distributed training and avoid port conflicts, or explicitly set `"env.strategy": "auto"` or `"env.strategy": "ddp_spawn"` to let AutoGluon choose a safe strategy. Alternatively, ensure that no other processes are using the required port, or specify a different port via the `"env.dist_port"` hyperparameter if multi-GPU is required and the environment supports it.

Node 10 (debug)

Property	Value
uct_value	1.4821
ctime	2026-05-21 17:52:44
debug_attempts	2
depth	3
execution_time	0.0
expected_child_count	0
failure_visits	1
id	10
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✓
num_children	0
prev_tutorial_prompt	### Condensed: Intro to Use Machine Learning or Deep Learning algorithms # Condensed: Intro to Use Machine Learning or Deep Learning algorithms Summary: You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you # Intro to Use Machine Learning or Deep Learning algorithms You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you have them
stage	debug
time_step	10
tool_used	machine learning
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.0
validated_visits	0

validation_score	None
visits	1
parent_id	4
child_ids	[]

Error Message:

```

stderr:
==> WARNING: A newer version of conda exists. <==
current version: 25.11.0
latest version: 26.5.0
Please update conda by running
$ conda update -n base -c conda-forge conda
warning: Requirements file `~/home/anri21/.conda/envs/mlzero_env/lib/python3.11/site-packages/autogluon/assistant/tools_registry/machine learning/requirements.txt`
does not contain any dependencies
Using Python 3.11.15 environment at: 44-addition_2/node_10/conda_env
Resolved 79 packages in 375ms
Uninstalled 1 package in 1.75s
Installed 78 packages in 36.63s
+ aiohappyeyeballs==2.6.2
+ aiohttp==3.13.5
+ aiosignal==1.4.0
+ annotated-doc==0.0.4
+ annotated-types==0.7.0
+ anyio==4.13.0
+ attrs==26.1.0
+ certifi==2026.5.20
+ chardet==7.4.3
+ charset-normalizer==3.4.7
+ click==8.4.0
+ cuda-bindings==13.2.0
+ cuda-pathfinder==1.5.4
+ cuda-toolkit==13.0.2
+ datasets==4.8.5
+ dill==0.4.1
+ filelock==3.29.0
+ frozenlist==1.8.0
+ fsspec==2026.2.0
+ h11==0.16.0
+ hf-xet==1.5.0
+ httpcore==1.0.9
+ httpx==0.28.1
+ huggingface-hub==1.16.0
+ idna==3.15

```

```
+ jinja2==3.1.6
+ markdown-it-py==4.2.0
+ markupsafe==3.0.3
+ mdurl==0.1.2
+ mpmath==1.3.0
+ multidict==6.7.1
+ multiprocessing==0.70.19
+ networkx==3.6.1
+ numpy==2.4.6
+ nvidia-cublas==13.1.1.3
+ nvidia-cuda-cupti==13.0.85
+ nvidia-cuda-nvrtc==13.0.88
+ nvidia-cuda-runtime==13.0.96
+ nvidia-cudnn-cu13==9.20.0.48
+ nvidia-cufft==12.0.0.61
+ nvidia-cufile==1.15.1.6
+ nvidia-curand==10.4.0.35
+ nvidia-cusolver==12.0.4.66
+ nvidia-cusparselt-cu13==0.8.1
+ nvidia-nccl-cu13==2.29.7
+ nvidia-nvjitlink==13.0.88
+ nvidia-nvshmem-cu13==3.4.5
+ nvidia-nvtx==13.0.85
+ opencv-python==4.13.0.92
+ pandas==3.0.3
+ pillow==12.2.0
+ propcache==0.5.2
+ pyarrow==24.0.0
+ pydantic==2.13.4
+ pydantic-core==2.46.4
+ pygments==2.20.0
+ python-calamine==0.6.2
+ python-dateutil==2.9.0.post0
+ pytz==2026.2
+ pyyaml==6.0.3
+ requests==2.34.2
+ rich==15.0.0
- setuptools==82.0.1
+ se... [truncated, 7681 chars total]
```

Error Analysis:

ERROR_SUMMARY: The error "DNN library initialization failed" with "Could not create cudnn handle: CUDNN_STATUS_NOT_INITIALIZED" indicates that TensorFlow was unable to initialize the cuDNN library required for GPU acceleration. This is typically caused by a mismatch between the installed NVIDIA driver, CUDA toolkit, and cuDNN versions, or by insufficient available GPU memory at initialization.

SUGGESTED_FIX:

Check that your NVIDIA driver version is compatible with the installed CUDA and cuDNN versions (driver 565.57.1 may not match CUDA 12.5/cuDNN 9.20). Ensure that the driver, CUDA toolkit, and cuDNN are all compatible with your TensorFlow version. Also, verify that sufficient GPU memory is available and no other processes are occupying the GPU. If the issue persists, try running TensorFlow in CPU mode to confirm the problem is GPU-specific.

Node 11 (debug)

Property	Value
uct_value	1.3384
ctime	2026-05-21 18:03:05
debug_attempts	3
depth	2
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	1.482079962591042
failure_penalty	-0.0
failure_visits	2
id	11
is_debug_successful	✗
is_leaf	✗
is_successful	✗
is_terminal	✓
normalized_failure_visit	0
num_children	1
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates using AutoGluon Multimodal for multimodal classification on the PetFinder dataset.
stage	debug
time_step	11
tool_used	autogluon.tabular
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.0

validated_visits	0
validation_score	None
visits	2
parent_id	2
child_ids	[14]

Error Message:

```

stderr:
==> WARNING: A newer version of conda exists. <==
current version: 25.11.0
latest version: 26.5.0
Please update conda by running
$ conda update -n base -c conda-forge conda
Using Python 3.11.15 environment at: 44-addition_2/node_11/conda_env
Resolved 239 packages in 1.25s
Uninstalled 1 package in 1.82s
Installed 237 packages in 1m 44s
+ absl-py==2.4.0
+ accelerate==1.13.0
+ adagio==0.2.6
+ aiohappyeyeballs==2.6.2
+ aiohttp==3.13.5
+ aiohttp-cors==0.8.1
+ aiosignal==1.4.0
+ alembic==1.18.4
+ annotated-doc==0.0.4
+ annotated-types==0.7.0
+ antlr4-python3-runtime==4.9.3
+ attrs==26.1.0
+ autogluon==1.5.1b20260521
+ autogluon-common==1.5.1b20260521
+ autogluon-core==1.5.1b20260521
+ autogluon-features==1.5.1b20260521
+ autogluon-multimodal==1.5.1b20260521
+ autogluon-tabular==1.5.1b20260521
+ autogluon-timeseries==1.5.1b20260521
+ beartype==0.22.9
+ beautifulsoup4==4.14.3
+ blis==1.3.3
+ boto3==1.43.12
+ botocore==1.43.12
+ catalogue==2.0.10

```

- + catboost==1.2.10
- + certifi==2026.5.20
- + cffi==2.0.0
- + chardet==7.4.3
- + charset-normalizer==3.4.7
- + chronos-forecasting==2.2.2
- + click==8.4.0
- + cloudpathlib==0.24.0
- + cloudpickle==3.1.2
- + colorama==0.4.6
- + colorful==0.6.0a1
- + colorlog==6.10.1
- + confection==1.3.3
- + contourpy==1.3.3
- + coreforecast==0.0.16
- + cryptography==48.0.0
- + cycler==0.12.1
- + cymem==2.0.13
- + daal==2025.9.0
- + datasets==4.0.0
- + defusedxml==0.7.1
- + dill==0.3.8
- + distlib==0.4.0
- + einops==0.8.2
- + einx==0.4.3
- + evaluate==0.4.6
- + fastai==2.8.5
- + fastcore==1.8.18
- + fastdownload==0.0.7
- + fastprogress==1.0.5
- + fasttransform==0.0.2
- + filelock==3.29.0
- + fonttools==4.63.0
- + frozendict==2.4.7
- + frozenlist==1.8.0
- + fsspec==2025.3.0
- + fugue==0.9.7
- + future==1.0.0
- + gdown==6.0.0
- + gluonts==0.16.2
- + google-api-core==2.30.3
- + google-auth==2.53.0
- + googleapis-common-protos==1.75.0
- + graphviz==0.21
- + grpcio==1.81.0rc1

+ hf-xet==1.5.0
+ httpx==1.0.dev3
+ huggingface-hub==0.36.2
+ hyperopt==0... [truncated, 174425 chars total]

Error Analysis:

ERROR_SUMMARY: The error occurred because the trained AutoGluon Tabular model expects the columns ['skin_tone', 'alternative_skin_tone', 'expert_opinion'] in the test data, but the provided test set only contains the 'image_name' column. As a result, a KeyError was raised during feature transformation, since the required features were missing from the test input.

SUGGESTED_FIX: Update your pipeline to use AutoGluon Multimodal (not Tabular) so that the model can learn directly from image files and any available tabular metadata. Specifically, ensure that during both training and inference, the test DataFrame includes a column with the absolute path to each image (e.g., 'image_path'), and configure the FeatureMetadata to mark this column as an 'image_path' type. If you want to use only image data for prediction (since test.csv has no metadata), train your model using only the image column and label, and ensure the test DataFrame passed to predict_proba contains the same image path column as used in training.

Node 12 (debug)

Property	Value
uct_value	1.4821
ctime	2026-05-21 18:13:56
debug_attempts	3
depth	3
execution_time	0.0
expected_child_count	0
failure_visits	1
id	12
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✓
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	debug
time_step	12
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.0
validated_visits	0
validation_score	None
visits	1
parent_id	3
child_ids	[]

Error Message:

stderr:

==> WARNING: A newer version of conda exists. <==

current version: 25.11.0

latest version: 26.5.0

Please update conda by running

\$ conda update -n base -c conda-forge conda

Using Python 3.11.15 environment at: 44-addition_2/node_12/conda_env

Resolved 235 packages in 2.31s

Uninstalled 1 package in 3.02s

Installed 233 packages in 1m 50s

+ absl-py==2.4.0

+ accelerate==1.13.0

+ adagio==0.2.6

+ aiohappyeyeballs==2.6.2

+ aiohttp==3.13.5

+ aiohttp-cors==0.8.1

+ aiosignal==1.4.0

+ alembic==1.18.4

+ annotated-doc==0.0.4

+ annotated-types==0.7.0

+ antlr4-python3-runtime==4.9.3

+ attrs==26.1.0

+ autogluon==1.5.1b20260521

+ autogluon-common==1.5.1b20260521

+ autogluon-core==1.5.1b20260521

+ autogluon-features==1.5.1b20260521

+ autogluon-multimodal==1.5.1b20260521

+ autogluon-tabular==1.5.1b20260521

+ autogluon-timeseries==1.5.1b20260521

+ beartype==0.22.9

+ beautifulsoup4==4.14.3

+ blis==1.3.3

+ boto3==1.43.12

+ botocore==1.43.12

+ catalogue==2.0.10

+ catboost==1.2.10

+ certifi==2026.5.20

+ cffi==2.0.0

+ chardet==7.4.3

+ charset-normalizer==3.4.7

+ chronos-forecasting==2.2.2

+ click==8.4.0

+ cloudpathlib==0.24.0

+ cloudpickle==3.1.2
+ colorama==0.4.6
+ colorful==0.6.0a1
+ colorlog==6.10.1
+ confection==1.3.3
+ contourpy==1.3.3
+ coreforecast==0.0.16
+ cryptography==48.0.0
+ cycler==0.12.1
+ cymem==2.0.13
+ datasets==4.0.0
+ defusedxml==0.7.1
+ dill==0.3.8
+ distlib==0.4.0
+ einops==0.8.2
+ einx==0.4.3
+ evaluate==0.4.6
+ fastai==2.8.5
+ fastcore==1.8.18
+ fastdownload==0.0.7
+ fastprogress==1.0.5
+ fasttransform==0.0.2
+ filelock==3.29.0
+ fonttools==4.63.0
+ frozendict==2.4.7
+ frozenlist==1.8.0
+ fsspec==2025.3.0
+ fugue==0.9.7
+ future==1.0.0
+ gdown==6.0.0
+ gluonts==0.16.2
+ google-api-core==2.30.3
+ google-auth==2.53.0
+ googleapis-common-protos==1.75.0
+ graphviz==0.21
+ grpcio==1.81.0rc1
+ hf-xet==1.5.0
+ httpx==1.0.dev3
+ huggingface-hub==0.36.2
+ hyperopt==0.2.7
+ idna==3.15... [truncated, 14785 chars total]

Error Analysis:

ERROR_SUMMARY: The error occurred during distributed training with PyTorch Lightning's DDP (Distributed Data Parallel) strategy. Specifically, torch.distributed.DistNetworkError was raised because the TCP port 20620 required for inter-process communication was already in use (EADDRINUSE), preventing

the distributed process group from initializing and causing the training to fail.

SUGGESTED_FIX:

Change the DDP communication port to a different, unused value by setting the environment variable `MASTER\_PORT` to a random high-numbered port (e.g., export MASTER_PORT=29500) before running your script. Alternatively, ensure that no other processes are occupying the default port, or switch to single-GPU training by setting "env.num_gpus": 1 in your AutoGluon hyperparameters if distributed training is not required.

Node 13 (debug)

Property	Value
uct_value	1.1772
ctime	2026-05-21 18:30:15
debug_attempts	3
depth	3
execution_time	0.0
expected_child_count	0
failure_visits	1
id	13
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✓
num_children	0
prev_tutorial_prompt	### Condensed: Intro to Use Machine Learning or Deep Learning algorithms # Condensed: Intro to Use Machine Learning or Deep Learning algorithms Summary: You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you # Intro to Use Machine Learning or Deep Learning algorithms You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you have them
stage	debug
time_step	13
tool_used	machine learning
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.0
validated_visits	0

validation_score	None
visits	1
parent_id	6
child_ids	[]

Error Message:

```

stderr:
==> WARNING: A newer version of conda exists. <==
current version: 25.11.0
latest version: 26.5.0
Please update conda by running
$ conda update -n base -c conda-forge conda
warning: Requirements file `~/home/anri21/.conda/envs/mlzero_env/lib/python3.11/site-packages/autogluon/assistant/tools_registry/machine learning/requirements.txt`
does not contain any dependencies
Using Python 3.11.15 environment at: 44-addition_2/node_13/conda_env
Resolved 79 packages in 721ms
Uninstalled 1 package in 4.32s
Installed 78 packages in 3m 03s
+ aiohappyeyeballs==2.6.2
+ aiohttp==3.13.5
+ aiosignal==1.4.0
+ annotated-doc==0.0.4
+ annotated-types==0.7.0
+ anyio==4.13.0
+ attrs==26.1.0
+ certifi==2026.5.20
+ chardet==7.4.3
+ charset-normalizer==3.4.7
+ click==8.4.0
+ cuda-bindings==13.2.0
+ cuda-pathfinder==1.5.4
+ cuda-toolkit==13.0.2
+ datasets==4.8.5
+ dill==0.4.1
+ filelock==3.29.0
+ frozenlist==1.8.0
+ fsspec==2026.2.0
+ h11==0.16.0
+ hf-xet==1.5.0
+ httpcore==1.0.9
+ httpx==0.28.1
+ huggingface-hub==1.16.0
+ idna==3.15

```

```
+ jinja2==3.1.6
+ markdown-it-py==4.2.0
+ markupsafe==3.0.3
+ mdurl==0.1.2
+ mpmath==1.3.0
+ multidict==6.7.1
+ multiprocessing==0.70.19
+ networkx==3.6.1
+ numpy==2.4.6
+ nvidia-cublas==13.1.1.3
+ nvidia-cuda-cupti==13.0.85
+ nvidia-cuda-nvrtc==13.0.88
+ nvidia-cuda-runtime==13.0.96
+ nvidia-cudnn-cu13==9.20.0.48
+ nvidia-cufft==12.0.0.61
+ nvidia-cufile==1.15.1.6
+ nvidia-curand==10.4.0.35
+ nvidia-cusolver==12.0.4.66
+ nvidia-cusparselt-cu13==0.8.1
+ nvidia-nccl-cu13==2.29.7
+ nvidia-nvjitlink==13.0.88
+ nvidia-nvshmem-cu13==3.4.5
+ nvidia-nvtx==13.0.85
+ opencv-python==4.13.0.92
+ pandas==3.0.3
+ pillow==12.2.0
+ propcache==0.5.2
+ pyarrow==24.0.0
+ pydantic==2.13.4
+ pydantic-core==2.46.4
+ pygments==2.20.0
+ python-calamine==0.6.2
+ python-dateutil==2.9.0.post0
+ pytz==2026.2
+ pyyaml==6.0.3
+ requests==2.34.2
+ rich==15.0.0
- setuptools==82.0.1
+ se... [truncated, 7725 chars total]
```

Error Analysis:

ERROR_SUMMARY: TensorFlow failed to initialize the cuDNN DNN library on the GPU, resulting in a runtime error during image feature extraction with EfficientNetB0. The error "Could not create cudnn handle: CUDNN_STATUS_NOT_INITIALIZED" and "DNN library initialization failed" indicates a mismatch or incompatibility between the installed CUDA/cuDNN libraries and the NVIDIA driver version, or a misconfiguration in the environment.

SUGGESTED_FIX:

Check that your NVIDIA driver, CUDA toolkit, and cuDNN versions are all compatible with the version of TensorFlow you are using. Ensure that the correct CUDA and cuDNN libraries are installed and visible in your environment (e.g., via `LD_LIBRARY_PATH`), and that no conflicting versions are present. If running in a managed environment, verify that the driver version (565.57.1) supports the installed CUDA/cuDNN versions, and consider downgrading or upgrading TensorFlow, CUDA, or cuDNN as needed to ensure compatibility.

Node 14 (debug)

Property	Value
uct_value	1.1772
ctime	2026-05-21 18:50:42
debug_attempts	3
depth	3
execution_time	0.0
expected_child_count	0
failure_visits	1
id	14
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✓
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates using AutoGluon Multimodal for multimodal classification on the PetFinder dataset.
stage	debug
time_step	14
tool_used	autogluon.tabular
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.0
validated_visits	0
validation_score	None
visits	1
parent_id	11
child_ids	[]

Error Message:

stderr:

==> WARNING: A newer version of conda exists. <==

current version: 25.11.0

latest version: 26.5.0

Please update conda by running

\$ conda update -n base -c conda-forge conda

Using Python 3.11.15 environment at: 44-addition_2/node_14/conda_env

Resolved 239 packages in 2.39s

Uninstalled 1 package in 3.40s

Installed 237 packages in 4m 18s

+ absl-py==2.4.0

+ accelerate==1.13.0

+ adagio==0.2.6

+ aiohappyeyeballs==2.6.2

+ aiohttp==3.13.5

+ aiohttp-cors==0.8.1

+ aiosignal==1.4.0

+ alembic==1.18.4

+ annotated-doc==0.0.4

+ annotated-types==0.7.0

+ antlr4-python3-runtime==4.9.3

+ attrs==26.1.0

+ autogluon==1.5.1b20260521

+ autogluon-common==1.5.1b20260521

+ autogluon-core==1.5.1b20260521

+ autogluon-features==1.5.1b20260521

+ autogluon-multimodal==1.5.1b20260521

+ autogluon-tabular==1.5.1b20260521

+ autogluon-timeseries==1.5.1b20260521

+ beartype==0.22.9

+ beautifulsoup4==4.14.3

+ blis==1.3.3

+ boto3==1.43.12

+ botocore==1.43.12

+ catalogue==2.0.10

+ catboost==1.2.10

+ certifi==2026.5.20

+ cffi==2.0.0

+ chardet==7.4.3

+ charset-normalizer==3.4.7

+ chronos-forecasting==2.2.2

+ click==8.4.0

+ cloudpathlib==0.24.0

```
+ cloudpickle==3.1.2
+ colorama==0.4.6
+ colorful==0.6.0a1
+ colorlog==6.10.1
+ confection==1.3.3
+ contourpy==1.3.3
+ coreforecast==0.0.16
+ cryptography==48.0.0
+ cycler==0.12.1
+ cymem==2.0.13
+ daal==2025.9.0
+ datasets==4.0.0
+ defusedxml==0.7.1
+ dill==0.3.8
+ distlib==0.4.0
+ einops==0.8.2
+ einx==0.4.3
+ evaluate==0.4.6
+ fastai==2.8.5
+ fastcore==1.8.18
+ fastdownload==0.0.7
+ fastprogress==1.0.5
+ fasttransform==0.0.2
+ filelock==3.29.0
+ fonttools==4.63.0
+ frozendict==2.4.7
+ frozenlist==1.8.0
+ fsspec==2025.3.0
+ fugue==0.9.7
+ future==1.0.0
+ gdown==6.0.0
+ gluonts==0.16.2
+ google-api-core==2.30.3
+ google-auth==2.53.0
+ googleapis-common-protos==1.75.0
+ graphviz==0.21
+ grpcio==1.81.0rc1
+ hf-xet==1.5.0
+ httpx==1.0.dev3
+ huggingface-hub==0.36.2
+ hyperopt==0... [truncated, 174902 chars total]
```

Error Analysis:

ERROR_SUMMARY: The error occurs because the test set provided to the trained AutoGluon TabularPredictor only contains the 'image_name' column, while the model expects the features 'skin_tone', 'alternative_skin_tone', and 'expert_opinion' (as used during training). As a result, a KeyError is raised when the predictor

attempts to transform the test data, since these required columns are missing.

SUGGESTED_FIX:

Update your prediction pipeline to ensure that the test data passed to `predictor.predict_proba` includes all feature columns used during training ('skin_tone', 'alternative_skin_tone', 'expert_opinion'), not just 'image_name'. If these features are not available for the test set, you must either (a) generate or impute reasonable values for them, or (b) switch to a modeling approach that does not require these features at inference (e.g., use only image data via a multimodal model as shown in the provided tutorials).

Node 15 (debug)

Property	Value
uct_value	1.4821
ctime	2026-05-21 19:07:14
debug_attempts	3
depth	3
execution_time	0.0
expected_child_count	0
failure_visits	1
id	15
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✓
num_children	0
prev_tutorial_prompt	### Condensed: Intro to Use Machine Learning or Deep Learning algorithms # Condensed: Intro to Use Machine Learning or Deep Learning algorithms Summary: You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you # Intro to Use Machine Learning or Deep Learning algorithms You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you have them
stage	debug
time_step	15
tool_used	machine learning
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.0
validated_visits	0

validation_score	None
visits	1
parent_id	4
child_ids	[]

Error Message:

```

stderr:
==> WARNING: A newer version of conda exists. <==
current version: 25.11.0
latest version: 26.5.0
Please update conda by running
$ conda update -n base -c conda-forge conda
warning: Requirements file `~/home/anri21/.conda/envs/mlzero_env/lib/python3.11/site-packages/autogluon/assistant/tools_registry/machine learning/requirements.txt`
does not contain any dependencies
Using Python 3.11.15 environment at: 44-addition_2/node_15/conda_env
Resolved 79 packages in 699ms
Uninstalled 1 package in 1.99s
Installed 78 packages in 52.41s
+ aiohappyeyeballs==2.6.2
+ aiohttp==3.13.5
+ aiosignal==1.4.0
+ annotated-doc==0.0.4
+ annotated-types==0.7.0
+ anyio==4.13.0
+ attrs==26.1.0
+ certifi==2026.5.20
+ chardet==7.4.3
+ charset-normalizer==3.4.7
+ click==8.4.0
+ cuda-bindings==13.2.0
+ cuda-pathfinder==1.5.4
+ cuda-toolkit==13.0.2
+ datasets==4.8.5
+ dill==0.4.1
+ filelock==3.29.0
+ frozenlist==1.8.0
+ fsspec==2026.2.0
+ h11==0.16.0
+ hf-xet==1.5.0
+ httpcore==1.0.9
+ httpx==0.28.1
+ huggingface-hub==1.16.0
+ idna==3.15

```

```
+ jinja2==3.1.6
+ markdown-it-py==4.2.0
+ markupsafe==3.0.3
+ mdurl==0.1.2
+ mpmath==1.3.0
+ multidict==6.7.1
+ multiprocessing==0.70.19
+ networkx==3.6.1
+ numpy==2.4.6
+ nvidia-cublas==13.1.1.3
+ nvidia-cuda-cupti==13.0.85
+ nvidia-cuda-nvrtc==13.0.88
+ nvidia-cuda-runtime==13.0.96
+ nvidia-cudnn-cu13==9.20.0.48
+ nvidia-cufft==12.0.0.61
+ nvidia-cufile==1.15.1.6
+ nvidia-curand==10.4.0.35
+ nvidia-cusolver==12.0.4.66
+ nvidia-cusparselt-cu13==0.8.1
+ nvidia-nccl-cu13==2.29.7
+ nvidia-nvjitlink==13.0.88
+ nvidia-nvshmem-cu13==3.4.5
+ nvidia-nvtx==13.0.85
+ opencv-python==4.13.0.92
+ pandas==3.0.3
+ pillow==12.2.0
+ propcache==0.5.2
+ pyarrow==24.0.0
+ pydantic==2.13.4
+ pydantic-core==2.46.4
+ pygments==2.20.0
+ python-calamine==0.6.2
+ python-dateutil==2.9.0.post0
+ pytz==2026.2
+ pyyaml==6.0.3
+ requests==2.34.2
+ rich==15.0.0
- setuptools==82.0.1
+ se... [truncated, 7708 chars total]
```

Error Analysis:

ERROR_SUMMARY: The error occurs during image feature extraction with TensorFlow/Keras EfficientNetB0, where the GPU/cuDNN initialization fails with "DNN library initialization failed" and "Could not create cudnn handle: CUDNN_STATUS_NOT_INITIALIZED." This is likely due to a mismatch or incompatibility between the installed NVIDIA driver, CUDA toolkit, and cuDNN versions, or insufficient GPU memory/resources, preventing TensorFlow from using the GPU for inference.

SUGGESTED_FIX:

Check that the NVIDIA driver version is compatible with the installed CUDA and cuDNN versions (e.g., driver $\geq$ CUDA runtime version), and ensure that the correct versions are installed in your conda environment. If running in a shared or resource-constrained environment, verify that sufficient GPU memory is available and no other processes are occupying the GPU; if necessary, try running TensorFlow in CPU mode by setting the environment variable `CUDA_VISIBLE_DEVICES=""` before script execution to bypass GPU/cuDNN issues.

Node 16 (debug)

Property	Value
uct_value	1.4821
ctime	2026-05-21 19:24:39
debug_attempts	2
depth	3
execution_time	0.0
expected_child_count	0
failure_visits	1
id	16
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✓
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates using AutoGluon Multimodal for multimodal classification on the PetFinder dataset.
stage	debug
time_step	16
tool_used	autogluon.tabular
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.0
validated_visits	0
validation_score	None
visits	1
parent_id	7
child_ids	[]

Error Message:

stderr:

==> WARNING: A newer version of conda exists. <==

current version: 25.11.0

latest version: 26.5.0

Please update conda by running

\$ conda update -n base -c conda-forge conda

Using Python 3.11.15 environment at: 44-addition_2/node_16/conda_env

Resolved 239 packages in 2.06s

Uninstalled 1 package in 2.89s

Installed 237 packages in 3m 51s

+ absl-py==2.4.0

+ accelerate==1.13.0

+ adagio==0.2.6

+ aiohappyeyeballs==2.6.2

+ aiohttp==3.13.5

+ aiohttp-cors==0.8.1

+ aiosignal==1.4.0

+ alembic==1.18.4

+ annotated-doc==0.0.4

+ annotated-types==0.7.0

+ antlr4-python3-runtime==4.9.3

+ attrs==26.1.0

+ autogluon==1.5.1b20260521

+ autogluon-common==1.5.1b20260521

+ autogluon-core==1.5.1b20260521

+ autogluon-features==1.5.1b20260521

+ autogluon-multimodal==1.5.1b20260521

+ autogluon-tabular==1.5.1b20260521

+ autogluon-timeseries==1.5.1b20260521

+ beartype==0.22.9

+ beautifulsoup4==4.14.3

+ blis==1.3.3

+ boto3==1.43.12

+ botocore==1.43.12

+ catalogue==2.0.10

+ catboost==1.2.10

+ certifi==2026.5.20

+ cffi==2.0.0

+ chardet==7.4.3

+ charset-normalizer==3.4.7

+ chronos-forecasting==2.2.2

+ click==8.4.0

+ cloudpathlib==0.24.0

+ cloudpickle==3.1.2
+ colorama==0.4.6
+ colorful==0.6.0a1
+ colorlog==6.10.1
+ confection==1.3.3
+ contourpy==1.3.3
+ coreforecast==0.0.16
+ cryptography==48.0.0
+ cycler==0.12.1
+ cymem==2.0.13
+ daal==2025.9.0
+ datasets==4.0.0
+ defusedxml==0.7.1
+ dill==0.3.8
+ distlib==0.4.0
+ einops==0.8.2
+ einx==0.4.3
+ evaluate==0.4.6
+ fastai==2.8.5
+ fastcore==1.8.18
+ fastdownload==0.0.7
+ fastprogress==1.0.5
+ fasttransform==0.0.2
+ filelock==3.29.0
+ fonttools==4.63.0
+ frozendict==2.4.7
+ frozenlist==1.8.0
+ fsspec==2025.3.0
+ fugue==0.9.7
+ future==1.0.0
+ gdown==6.0.0
+ gluonts==0.16.2
+ google-api-core==2.30.3
+ google-auth==2.53.0
+ googleapis-common-protos==1.75.0
+ graphviz==0.21
+ grpcio==1.81.0rc1
+ hf-xet==1.5.0
+ httpx==1.0.dev3
+ huggingface-hub==0.36.2
+ hyperopt==0... [truncated, 174988 chars total]

Error Analysis:

ERROR_SUMMARY: The error occurred because the test set provided to the trained AutoGluon TabularPredictor was missing required columns ('skin_tone', 'alternative_skin_tone', 'expert_opinion') that were present during training. As a result, AutoGluon raised a KeyError when attempting to transform the test data, since it

expects the same feature columns as in the training data.

SUGGESTED_FIX:

Ensure that the test set DataFrame passed to ``predictor.predict_proba()`` contains all the feature columns used during training, specifically 'skin_tone', 'alternative_skin_tone', and 'expert_opinion', in addition to 'image_name'. Merge or join these metadata columns from the training set or another source into the test set before making predictions, so the test DataFrame matches the training feature schema.

Node 17 (debug)

Property	Value
uct_value	1.4821
ctime	2026-05-21 19:44:36
debug_attempts	3
depth	3
execution_time	0.0
expected_child_count	0
failure_visits	1
id	17
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✓
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates using AutoGluon Multimodal for multimodal classification on the PetFinder dataset.
stage	debug
time_step	17
tool_used	autogluon.tabular
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.0
validated_visits	0
validation_score	None
visits	1
parent_id	7
child_ids	[]

Error Message:

stderr:

==> WARNING: A newer version of conda exists. <==

current version: 25.11.0

latest version: 26.5.0

Please update conda by running

\$ conda update -n base -c conda-forge conda

Using Python 3.11.15 environment at: 44-addition_2/node_17/conda_env

Resolved 239 packages in 4.59s

Uninstalled 1 package in 2.48s

Installed 237 packages in 3m 05s

+ absl-py==2.4.0

+ accelerate==1.13.0

+ adagio==0.2.6

+ aiohappyeyeballs==2.6.2

+ aiohttp==3.13.5

+ aiohttp-cors==0.8.1

+ aiosignal==1.4.0

+ alembic==1.18.4

+ annotated-doc==0.0.4

+ annotated-types==0.7.0

+ antlr4-python3-runtime==4.9.3

+ attrs==26.1.0

+ autogluon==1.5.1b20260521

+ autogluon-common==1.5.1b20260521

+ autogluon-core==1.5.1b20260521

+ autogluon-features==1.5.1b20260521

+ autogluon-multimodal==1.5.1b20260521

+ autogluon-tabular==1.5.1b20260521

+ autogluon-timeseries==1.5.1b20260521

+ beartype==0.22.9

+ beautifulsoup4==4.14.3

+ blis==1.3.3

+ boto3==1.43.12

+ botocore==1.43.12

+ catalogue==2.0.10

+ catboost==1.2.10

+ certifi==2026.5.20

+ cffi==2.0.0

+ chardet==7.4.3

+ charset-normalizer==3.4.7

+ chronos-forecasting==2.2.2

+ click==8.4.0

+ cloudpathlib==0.24.0

```
+ cloudpickle==3.1.2
+ colorama==0.4.6
+ colorful==0.6.0a1
+ colorlog==6.10.1
+ confection==1.3.3
+ contourpy==1.3.3
+ coreforecast==0.0.16
+ cryptography==48.0.0
+ cycler==0.12.1
+ cymem==2.0.13
+ daal==2025.9.0
+ datasets==4.0.0
+ defusedxml==0.7.1
+ dill==0.3.8
+ distlib==0.4.0
+ einops==0.8.2
+ einx==0.4.3
+ evaluate==0.4.6
+ fastai==2.8.5
+ fastcore==1.8.18
+ fastdownload==0.0.7
+ fastprogress==1.0.5
+ fasttransform==0.0.2
+ filelock==3.29.0
+ fonttools==4.63.0
+ frozendict==2.4.7
+ frozenlist==1.8.0
+ fsspec==2025.3.0
+ fugue==0.9.7
+ future==1.0.0
+ gdown==6.0.0
+ gluonts==0.16.2
+ google-api-core==2.30.3
+ google-auth==2.53.0
+ googleapis-common-protos==1.75.0
+ graphviz==0.21
+ grpcio==1.81.0rc1
+ hf-xet==1.5.0
+ httpx==1.0.dev3
+ huggingface-hub==0.36.2
+ hyperopt==0... [truncated, 174151 chars total]
```

Error Analysis:

ERROR_SUMMARY: The error occurs because the trained AutoGluon Tabular model expects the columns 'skin_tone', 'alternative_skin_tone', and 'expert_opinion' in the test set, but the provided test.csv only contains 'image_name'. As a result, when calling `predict\_proba`, AutoGluon raises a KeyError since the required features

are missing from the test data.

SUGGESTED_FIX:

Modify your inference pipeline to ensure that the test DataFrame passed to ``predict_proba`` contains all columns used during training, including 'skin_tone', 'alternative_skin_tone', and 'expert_opinion'. If these features are unavailable for test images, you must impute them with default or placeholder values (e.g., using the mode or a special value such as -1 or NaN) so that the test DataFrame matches the training feature set in both column names and types before making predictions.